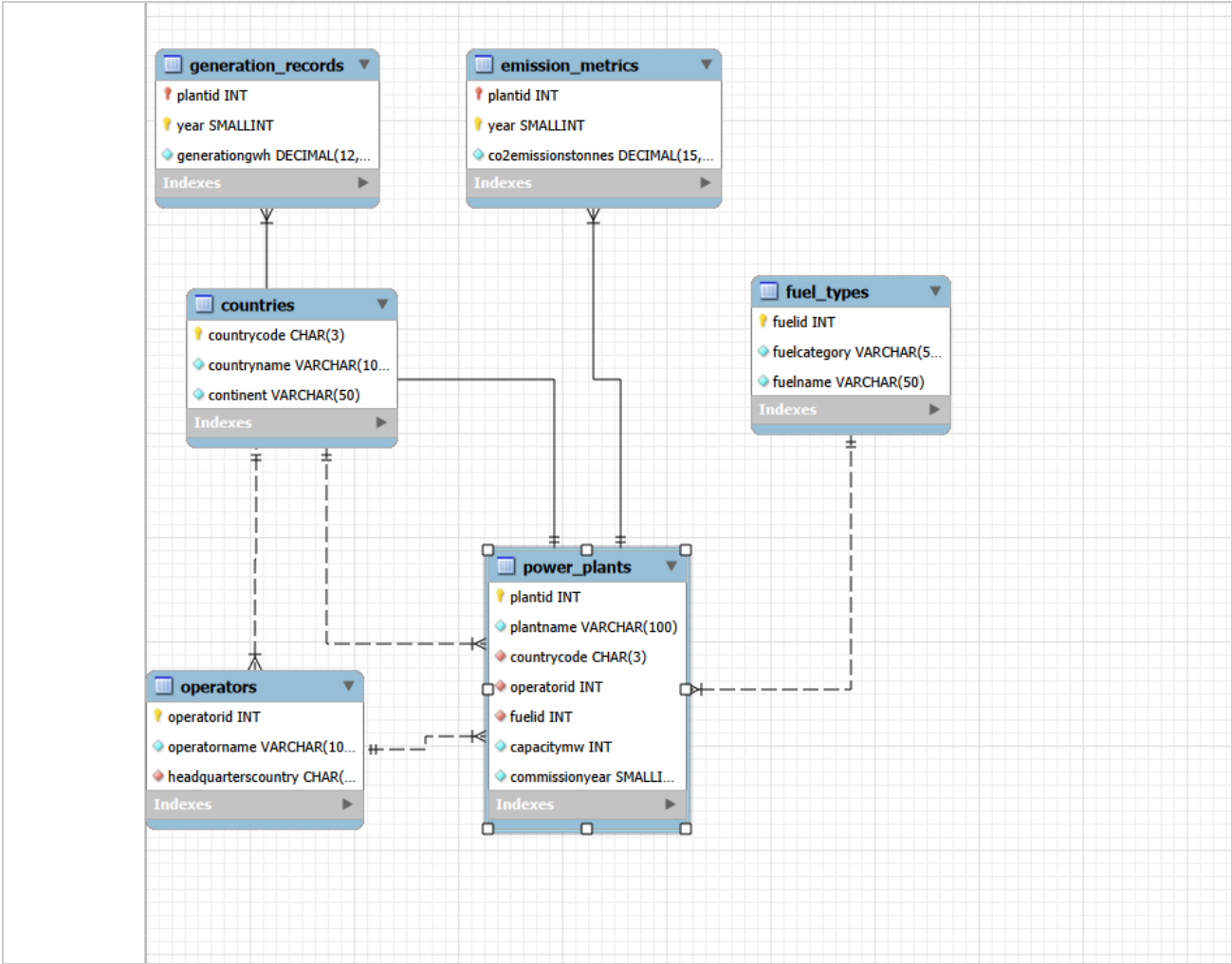


Assignment 3 (HW3)

Global Energy and Power Plants Relational Model in MySQL

EER Diagram

Reverse-engineered in MySQL Workbench from the assignment3_powerplant schema. Primary keys, foreign keys, the composite keys (plantid, year) on generation_records and emission_metrics, and all relationships are shown.



Query 1

Retrieve the plant name, country name, operator name, fuel category, fuel name, capacity (in MW), and commission year for all power plants without using table aliases. Sort the results in descending order by capacity.

SQL

```
select plantname, countryname, operatorname, fuelcategory, fuelname, capacitymw,
commissionyear
from power_plants
    inner join countries on power_plants.countrycode = countries.countrycode
    inner join operators on power_plants.operatorid = operators.operatorid
    inner join fuel_types on power_plants.fuelid = fuel_types.fuelid
order by capacitymw desc;
```

Result (MySQL Workbench)

Result Grid





Filter Rows:

Export:



Wrap Cell Content:



	plantname	countryname	operatorname	fuelcategory	fuelname	capacitymw	commissionyear
▶	Three Gorges	China	China Yangtze Power	Renewable	Hydro	22500	2003
	Itaipu	Brazil	Electrobras	Renewable	Hydro	14000	1984
	Tuocketuo	China	China Yangtze Power	Fossil	Coal	13200	1995
	Kashiwazaki-Kariwa	Japan	Tepco Energy	Nuclear	Uranium	8212	1985
	Grand Coulee	United States	Global Hydro Inc	Renewable	Hydro	6809	1942
	Vindhyachal	India	NTPC Limited	Fossil	Coal	4760	1987
	Janschwalde	Germany	Rhein Power AG	Fossil	Coal	3000	1981
	Loy Yang	Australia	Snowy Hydro	Fossil	Coal	2210	1984

Query 2

Retrieve the plant name, country code, calendar year, and annual generation (in GWh) for all power plants for the year 2024 without using table aliases. Sort the results in descending order by generation.

SQL

```
select plantname, countrycode, year, generationgwh
from power_plants
    -- inner join countries on power_plants.countrycode = countries.countrycode
(already done above!)
    inner join generation_records on power_plants.plantid =
generation_records.plantid
where year = 2024
order by generationgwh desc;
```

Result (MySQL Workbench)

plantname	countrycode	year	generationgwh
Three Gorges	CHN	2024	98800.00
Itaipu	BRA	2024	89000.10
Tuocketuo	CHN	2024	71000.30
Vindhyachal	IND	2024	35600.80
Grand Coulee	USA	2024	21500.50
Janschwalde	DEU	2024	18400.20
Loy Yang	AUS	2024	14200.40
Kashiwazaki-Kariwa	JPN	2024	0.00

Query 3

Retrieve the plant name, country code, calendar year, annual generation (in GWh), and CO2 emissions (in tonnes) for all power plants for the year 2024 without using table aliases. Sort the results in ascending order by CO2 emissions.

SQL

```
select plantname, countrycode, generation_records.year, generationgwh, co2emissionstonnes
from power_plants
      inner join generation_records on power_plants.plantid =
generation_records.plantid
      inner join emission_metrics on generation_records.plantid =
emission_metrics.plantid
                                and generation_records.year = emission_metrics.year
where generation_records.year = 2024
order by co2emissionstonnes;
```

Result (MySQL Workbench)

	plantname	countrycode	year	generationgwh	co2emissionstonnes
►	Grand Coulee	USA	2024	21500.50	0.00
	Itaipu	BRA	2024	89000.10	0.00
	Kashiwazaki-Kariwa	JPN	2024	0.00	0.00
	Three Gorges	CHN	2024	98800.00	0.00
	Loy Yang	AUS	2024	14200.40	1950000.00
	Janschwalde	DEU	2024	18400.20	2450000.00
	Vindhyachal	IND	2024	35600.80	3100000.00
	Tuoketuo	CHN	2024	71000.30	5200000.00

Query 4

Write a SQL query using a Common Table Expression (CTE) to calculate the total cumulative power generation (in GWh) for each operator across all available years without using table aliases. Retrieve the operator name, headquarters country, and their total generated power, and sort the results in descending order by total generation.

SQL

```
with operator_totals as (  
    select operatorid, sum(generationgwh) as total_generation  
    from power_plants  
    inner join generation_records on power_plants.plantid = generation_records.plantid  
    group by operatorid  
)  
select operatorname, headquarterscountry, total_generation  
from operators  
inner join operator_totals on operators.operatorid = operator_totals.operatorid  
order by total_generation desc;
```

Result (MySQL Workbench)

operatorname	headquarterscountry	total_generation
China Yangtze Power	CHN	169800.30
Electrobras	BRA	89000.10
NTPC Limited	IND	35600.80
Global Hydro Inc	USA	21500.50
Rhein Power AG	DEU	18400.20
Snowy Hydro	AUS	14200.40
Tepco Energy	JPN	0.00

Query 5

Write a SQL query using two separate Common Table Expressions (CTEs)--one to calculate total power generation by country code and another to calculate total CO2 emissions by country code without using table aliases. Then, join these CTEs with the countries table to display the country name, total generation (in GWh), and total CO2 emissions (in tonnes). Sort the results in descending order by total generation.

SQL

```
with total_power_generation as (  
    select countrycode, sum(generationgwh) as total_generation  
    from power_plants  
        inner join generation_records on power_plants.plantid =  
generation_records.plantid  
    group by countrycode  
) ,  
total_co2_generation as (  
    select countrycode, sum(co2emissionstonnes) as total_emission  
    from power_plants  
        inner join emission_metrics on power_plants.plantid =  
emission_metrics.plantid  
    group by countrycode  
)  
  
select countryname, total_generation, total_emission  
from countries  
        inner join total_power_generation on countries.countrycode =  
total_power_generation.countrycode  
        inner join total_co2_generation on countries.countrycode =  
total_co2_generation.countrycode  
order by total_generation desc;
```

Result (MySQL Workbench)

countryname	total_generation	total_emission
China	169800.30	5200000.00
Brazil	89000.10	0.00
India	35600.80	3100000.00
United States	21500.50	0.00
Germany	18400.20	2450000.00
Australia	14200.40	1950000.00
Japan	0.00	0.00